

Agenda - Scraping Top Gainer 25 CryptoCurrencies

- Importing Libraries
- Adding Headers
- Requesting url
- Getting Response
- Accessing the Code
- Finding Table and its Rows that Contains the Details
- Creating Empty List
- Accessing Required Information
- Naming Columns
- Converting Into DataFrame
- Saving to CSV File
- Number of Rows and Columns
- Using Describe Function
- Using Sample Function
- Using Head Function
- Checking null values sum

Scraping Yahoo! Finance

Top 25 Gainer Crypto Currency

Importing Libraries

In [1]:
```
import requests
from bs4 import BeautifulSoup
import pandas as pd
```

Adding Headers

In [2]:
```
headers = {
    "User-Agent": "Mozilla/5.0"
}
```

Requesting url

In [3]:
```
url = "https://finance.yahoo.com/markets/crypto/gainers/"
```

In [4]:
```
response = requests.get(url, headers=headers)
```

Accessing the Code

In [5]:
```
soup = BeautifulSoup(response.text, "html.parser")
```

Finding Table and its Rows that Contains the Details

In [6]:
```
table = soup.find("table")
if table is None:
    print("Table not found. The data may be loaded dynamically")
else:
```

```
    rows = table.find("tbody").find_all("tr")
```

Creating Empty List

In [7]:
```
data = []
```

Accessing required Information

In [8]:
```
for row in rows:
        cols = row.find_all("td")
        data.append([col.get_text(strip=True) for col in cols])
```

In [9]:
```
data
```

Out[9]:
```
[['PORT11220-USD',
  'Port Finance USD',
  '',
  '0.004388+0.004231(+2,694.87%)',
  '+0.004231',
  '+2,694.87%',
  '0',
  '82,729',
  '82,729',
  '82,729',
  '0',
  '-97.89%',
  ''],
 ['CATX31291-USD',
  'CATX USD',
  '',
  '0.00000636+0.00000608(+2,168.12%)',
  '+0.00000608',
  '+2,168.12%',
  '0',
  '122,282',
  '122,282',
  '122,282',
  '0',
  '+600.00%',
  ''],
 ['SNL35953-USD',
  'Snow Leopard USD',
  '',
  '0.00000032+0.00000029(+1,055.80%)',
  '+0.00000029',
  '+1,055.80%',
  '0',
  '131,669',
  '131,669',
  '131,669',
  '0',
  '--',
  ''],
 ['MAGA33521-USD',
  'PEPE MAGA USD',
  '',
  '0.002745+0.002387(+666.49%)',
  '+0.002387',
  '+666.49%',
  '0',
  '930,745',
  '930,745',
  '930,745',
  '0',
  '+19,050.00%',
  ''],
 ['NEIRO33184-USD',
  'Neiro Pump USD',
  '',
  '0.001007+0.000856(+567.29%)',
  '+0.000856',
  '+567.29%',
  '0',
  '499,873',
  '499,873',
  '499,873',
  '0',
  '+9,900.00%',
  ''],
 ['DARAM-USD',
  'DARAM AI USD',
```

```
 '',
 '0.00000009+0.00000007(+477.74%)',
 '+0.00000007',
 '+477.74%',
 '0',
 '579,884',
 '579,884',
 '579,884',
 '0',
 '--',
 ''],
['LOVELY-USD',
 'Lovely Finance [Old] USD',
 '',
 '0.00000433+0.00000360(+488.84%)',
 '+0.00000360',
 '+488.84%',
 '0',
 '44,497',
 '44,497',
 '44,497',
 '0',
 '-100.00%',
 ''],
['TURBO36554-USD',
 'TURBO BOME USD',
 '',
 '0.00000789+0.00000651(+469.60%)',
 '+0.00000651',
 '+469.60%',
 '0',
 '514,969',
 '514,969',
 '514,969',
 '0',
 '+23.53%',
 ''],
['BOME33774-USD',
 'BOME AI USD',
 '',
 '0.00000021+0.00000016(+355.43%)',
 '+0.00000016',
 '+355.43%',
 '0',
 '694,896',
 '694,896',
 '694,896',
 '0',
 '--',
 ''],
['TRUMP33561-USD',
 'trumpwifhat USD',
 '',
 '0.016647+0.013494(+428.05%)',
 '+0.013494',
 '+428.05%',
 '0',
 '602,549',
 '602,549',
 '602,549',
 '0',
 '+21,162.50%',
 ''],
['TURBO33724-USD',
 'Turbo Trump USD',
 '',
 '0.000851+0.000717(+533.18%)',
 '+0.000717',
 '+533.18%',
 '0',
 '4.564M',
 '4.564M',
 '4.564M',
 '0',
 '+17,600.00%',
 ''],
['MSTR34245-USD',
 'MSTR USD',
 '',
 '0.850634+0.687558(+421.62%)',
 '+0.687558',
 '+421.62%',
 '0',
```

```
 '11.108M',
 '11.108M',
 '11.108M',
 '0',
 '+15,911.19%',
 ''],
['GROK33471-USD',
 'SORA GROK USD',
 '',
 '0.015001+0.012085(+414.41%)',
 '+0.012085',
 '+414.41%',
 '0',
 '5.037M',
 '5.037M',
 '5.037M',
 '0',
 '+1,420,344.34%',
 ''],
['MAGA32751-USD',
 'TRUMP MAGA (trumpmaga.me) USD',
 '',
 '0.014314+0.011449(+399.54%)',
 '+0.011449',
 '+399.54%',
 '0',
 '819,565',
 '819,565',
 '819,565',
 '0',
 '+11,785.71%',
 ''],
['SQUID34407-USD',
 'Squid Game (squidgame.top) USD',
 '',
 '0.000230+0.000182(+385.23%)',
 '+0.000182',
 '+385.23%',
 '0',
 '710,171',
 '710,171',
 '710,171',
 '0',
 '+600.00%',
 ''],
['CAT32486-USD',
 'NOT USD',
 '',
 '0.000028+0.000020(+272.69%)',
 '+0.000020',
 '+272.69%',
 '0',
 '2.686M',
 '2.686M',
 '2.686M',
 '0',
 '+2,172.22%',
 ''],
['STOCK37174-USD',
 'Coin Stock USD',
 '',
 '20.25+13.54(+201.61%)',
 '+13.54',
 '+201.61%',
 '0',
 '8.132M',
 '8.132M',
 '8.132M',
 '0',
 '-93.53%',
 ''],
['GUD-USD',
 'Gud Tech USD',
 '',
 '0.000993+0.000748(+304.85%)',
 '+0.000748',
 '+304.85%',
 '0',
 '64,450',
 '64,450',
 '64,450',
 '0',
 '-98.94%',
```

```
 ''],
['TSLA37391-USD',
 'Tesla USD',
 '',
 '65.79+50.16(+320.79%)',
 '+50.16',
 '+320.79%',
 '0',
 '2.595M',
 '2.595M',
 '2.595M',
 '0',
 '-82.78%',
 ''],
['HOOD37295-USD',
 'Robinhood USD',
 '',
 '0.050586+0.040128(+383.70%)',
 '+0.040128',
 '+383.70%',
 '0',
 '3.484M',
 '3.484M',
 '3.484M',
 '0',
 '+1,213.63%',
 ''],
['BITCOIN33683-USD',
 'HarryPotterObamaSonic10Inu 2.0 USD',
 '',
 '0.073898+0.052804(+250.34%)',
 '+0.052804',
 '+250.34%',
 '0',
 '132,949',
 '132,949',
 '132,949',
 '0',
 '+609,200.00%',
 ''],
['BROCCOLI35787-USD',
 'Broccoli(broccolicoin.me) USD',
 '',
 '0.001506+0.001107(+277.80%)',
 '+0.001107',
 '+277.80%',
 '0',
 '2.891M',
 '2.891M',
 '2.891M',
 '0',
 '-85.73%',
 ''],
['PEOPLE32079-USD',
 'MAGA USD',
 '',
 '0.000308+0.000222(+257.94%)',
 '+0.000222',
 '+257.94%',
 '0',
 '1.671M',
 '1.671M',
 '1.671M',
 '0',
 '+290.91%',
 ''],
['GME32488-USD',
 'GameStop Coin USD',
 '',
 '0.000214+0.000156(+267.15%)',
 '+0.000156',
 '+267.15%',
 '0',
 '2.566M',
 '2.566M',
 '2.566M',
 '0',
 '+866.67%',
 ''],
['DOGS32713-USD',
 'HARRIS DOGS USD',
 '',
 '0.000060+0.000043(+254.69%)',
```

```
    '+0.000043',
    '+254.69%',
    '0',
    '2.527M',
    '2.527M',
    '2.527M',
    '0',
    '--',
    '']]
```

In [10]:
```
print(data[0])
print(len(data[0]))
```

```
['PORT11220-USD', 'Port Finance USD', '', '0.004388+0.004231(+2,694.87%)', '+0.004231', '+2,694.87%', '0', '82,729', '82,729', '82,729', '0', '-97.89%', '']
13
```

Naming Columns

In [11]:
```
columns=[
        "Symbol",
        "Name",
        " ",
        "Price",
        "Change",
        "% Change",
        "Market Cap",
        "Volume ",
        "Volume in currency( 24 hr)",
        "Volume in all currencies(in 24 hr)",
        "Circulating Supply",
        "52 Weeks Change%",
        " "
]
```

Converting into DataFrame

In [12]:
```
df = pd.DataFrame(data, columns=columns)
```

In [13]:
```
df
```

Out[13]:

	Symbol	Name	Price	Change	% Change	Market Cap	Volume
0	PORT11220-USD	Port Finance USD	0.004388+0.004231(+2,694.87%)	+0.004231	+2,694.87%	0	82,729
1	CATX31291-USD	CATX USD	0.00000636+0.00000608(+2,168.12%)	+0.00000608	+2,168.12%	0	122,282
2	SNL35953-USD	Snow Leopard USD	0.00000032+0.00000029(+1,055.80%)	+0.00000029	+1,055.80%	0	131,669
3	MAGA33521-USD	PEPE MAGA USD	0.002745+0.002387(+666.49%)	+0.002387	+666.49%	0	930,745
4	NEIRO33184-USD	Neiro Pump USD	0.001007+0.000856(+567.29%)	+0.000856	+567.29%	0	499,873
5	DARAM-USD	DARAM AI USD	0.00000009+0.00000007(+477.74%)	+0.00000007	+477.74%	0	579,884
6	LOVELY-USD	Lovely Finance [Old] USD	0.00000433+0.00000360(+488.84%)	+0.00000360	+488.84%	0	44,497
7	TURBO36554-USD	TURBO BOME USD	0.00000789+0.00000651(+469.60%)	+0.00000651	+469.60%	0	514,969
8	BOME33774-USD	BOME AI USD	0.00000021+0.00000016(+355.43%)	+0.00000016	+355.43%	0	694,896
9	TRUMP33561-USD	trumpwifhat USD	0.016647+0.013494(+428.05%)	+0.013494	+428.05%	0	602,549
10	TURBO33724-USD	Turbo Trump USD	0.000851+0.000717(+533.18%)	+0.000717	+533.18%	0	4.564M
11	MSTR34245-USD	MSTR USD	0.850634+0.687558(+421.62%)	+0.687558	+421.62%	0	11.108M
12	GROK33471-USD	SORA GROK USD	0.015001+0.012085(+414.41%)	+0.012085	+414.41%	0	5.037M
13	MAGA32751-USD	TRUMP MAGA (trumpmaga.me) USD	0.014314+0.011449(+399.54%)	+0.011449	+399.54%	0	819,565
14	SQUID34407-USD	Squid Game (squidgame.top) USD	0.000230+0.000182(+385.23%)	+0.000182	+385.23%	0	710,171
15	CAT32486-USD	NOT USD	0.000028+0.000020(+272.69%)	+0.000020	+272.69%	0	2.686M
16	STOCK37174-USD	Coin Stock USD	20.25+13.54(+201.61%)	+13.54	+201.61%	0	8.132M
17	GUD-USD	Gud Tech USD	0.000993+0.000748(+304.85%)	+0.000748	+304.85%	0	64,450
18	TSLA37391-USD	Tesla USD	65.79+50.16(+320.79%)	+50.16	+320.79%	0	2.595M
19	HOOD37295-USD	Robinhood USD	0.050586+0.040128(+383.70%)	+0.040128	+383.70%	0	3.484M
20	BITCOIN33683-USD	HarryPotterObamaSonic10Inu 2.0 USD	0.073898+0.052804(+250.34%)	+0.052804	+250.34%	0	132,949
21	BROCCOLI35787-USD	Broccoli(broccolicoin.me) USD	0.001506+0.001107(+277.80%)	+0.001107	+277.80%	0	2.891M
22	PEOPLE32079-USD	MAGA USD	0.000308+0.000222(+257.94%)	+0.000222	+257.94%	0	1.671M
23	GME32488-USD	GameStop Coin USD	0.000214+0.000156(+267.15%)	+0.000156	+267.15%	0	2.566M
24	DOGS32713-USD	HARRIS DOGS USD	0.000060+0.000043(+254.69%)	+0.000043	+254.69%	0	2.527M

Saving to CSV file

In [14]:
```
df.to_csv("Top Gainer Crypto currency.csv")
```

Number of Rows and Columns

In [15]:
```
df.shape
```

Out[15]: (25, 13)

Using Describe Function

In [16]:
```
df.describe()
```

Out[16]:

	Symbol	Name		Price	Change	% Change	Market Cap	Volume	Volume in currency(24 hr)	Volume in all currencies(in 24 hr)
count	25	25	25	25	25	25	25	25	25	25
unique	25	25	1	25	25	25	1	25	25	25
top	PORT11220-USD	Port Finance USD		0.004388+0.004231(+2,694.87%)	+0.004231	+2,694.87%	0	82,729	82,729	82,729
freq	1	1	25	1	1	1	25	1	1	1

Using Sample Function

In [17]:
```
df.sample(10)
```

Out[17]:

	Symbol	Name	Price	Change	% Change	Market Cap	Volume	Volume in currency(24 hr)	Volume currenci 2
1	CATX31291-USD	CATX USD	0.00000636+0.00000608(+2,168.12%)	+0.00000608	+2,168.12%	0	122,282	122,282	12
12	GROK33471-USD	SORA GROK USD	0.015001+0.012085(+414.41%)	+0.012085	+414.41%	0	5.037M	5.037M	5.
18	TSLA37391-USD	Tesla USD	65.79+50.16(+320.79%)	+50.16	+320.79%	0	2.595M	2.595M	2.
6	LOVELY-USD	Lovely Finance [Old] USD	0.00000433+0.00000360(+488.84%)	+0.00000360	+488.84%	0	44,497	44,497	4
11	MSTR34245-USD	MSTR USD	0.850634+0.687558(+421.62%)	+0.687558	+421.62%	0	11.108M	11.108M	11.
5	DARAM-USD	DARAM AI USD	0.00000009+0.00000007(+477.74%)	+0.00000007	+477.74%	0	579,884	579,884	57
9	TRUMP33561-USD	trumpwifhat USD	0.016647+0.013494(+428.05%)	+0.013494	+428.05%	0	602,549	602,549	60
3	MAGA33521-USD	PEPE MAGA USD	0.002745+0.002387(+666.49%)	+0.002387	+666.49%	0	930,745	930,745	93
19	HOOD37295-USD	Robinhood USD	0.050586+0.040128(+383.70%)	+0.040128	+383.70%	0	3.484M	3.484M	3.
16	STOCK37174-USD	Coin Stock USD	20.25+13.54(+201.61%)	+13.54	+201.61%	0	8.132M	8.132M	8.

Using Head Function

In [18]:
```
df.head()
```

Out[18]:

	Symbol	Name	Price	Change	% Change	Market Cap	Volume	Volume in currency(24 hr)	Volume in all currencies(in 24 hr)
0	PORT11220-USD	Port Finance USD	0.004388+0.004231(+2,694.87%)	+0.004231	+2,694.87%	0	82,729	82,729	82,729
1	CATX31291-USD	CATX USD	0.00000636+0.00000608(+2,168.12%)	+0.00000608	+2,168.12%	0	122,282	122,282	122,282
2	SNL35953-USD	Snow Leopard USD	0.00000032+0.00000029(+1,055.80%)	+0.00000029	+1,055.80%	0	131,669	131,669	131,669
3	MAGA33521-USD	PEPE MAGA USD	0.002745+0.002387(+666.49%)	+0.002387	+666.49%	0	930,745	930,745	930,745
4	NEIRO33184-USD	Neiro Pump USD	0.001007+0.000856(+567.29%)	+0.000856	+567.29%	0	499,873	499,873	499,873

Checking null values sum

In [19]:
```
df.isnull().sum()
```

Out[19]:
```
Symbol                                  0
Name                                    0
                                        0
Price                                   0
Change                                  0
% Change                                0
Market Cap                              0
Volume                                  0
Volume in currency( 24 hr)              0
Volume in all currencies(in 24 hr)      0
Circulating Supply                      0
52 Weeks Change%                        0
                                        0
dtype: int64
```

End



In [ ]: